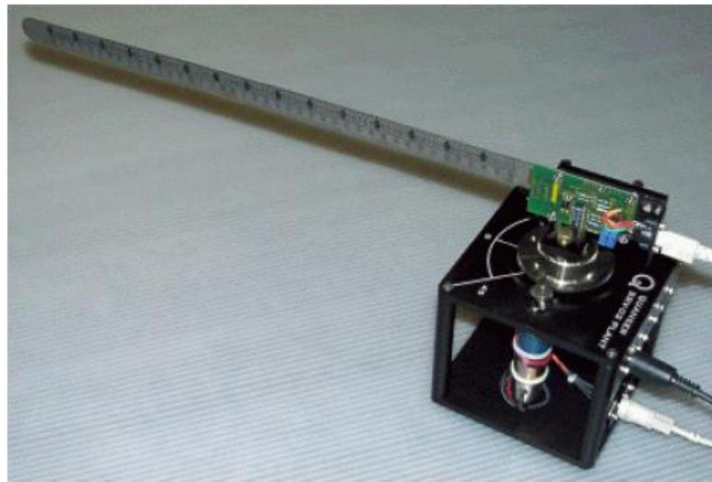


Control of Cyber-physical Systems

LABORATORY WORK REPORT — PART II

Computer Control of a Flexible Robot Arm Joint



Shift L04

Group 21:

Bernardo Lopes — 102559

Diogo Sousa — 107004

Beatriz Coelho — 107113

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Instituto Superior Técnico
Department of Electrical and Computer Engineering
Scientific Area of Systems, Decision, and Control

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1 Question 4 - Controller design and testing of the controlled system (17 pts)

1.1 Insert the MATLAB scripts, comments included. (1.5 pts)

```
1 %% Load matrices
2 load("SS_4_Model.mat"); %Load the file with the identified model of
  report 1
3 A = sys.A; B = sys.B; C = sys.C; D = sys.D; Ts = 0.02; %Assign the
  matrices
4 G = ss(A, B, C, 0, Ts); %Transform the matrices into a MATLAB state-
  space
5
6 Kp = -34.5846; Ke = 1.9213; % Sensor Calibration constants
7 K_cal = [Kp Ke 0 0]; % Gain to compute the tip angle from the output
  signals of the plant
8
9 %% Compute the Nu and Nx values to allow for the introduction of a
  reference
10 % According to (8.73) Franklin DCDS 3rd Ed
11 N = inv([A-eye(size(A)), B; C,0])*[zeros(size(A,1),1);1];
12 Nx = N(1:end-1,:);
13 Nu = N(end,:);
14
15 %% Calculate LQR vector gain and compensation gain
16 Q = C'*C;
17 R = 400;
18 [K, ~, E] = dlqr(A, B, Q, R);
19 Nbar = Nu + K * Nx;
20
21 %% Compute LQE vector gain for the chosen covariance matrices
22 w2 = 100; QE = eye(size(A))*w2;
23 RE = 1;
24 G1 = eye(size(A));
25 [M,P,Z,EE] = dlqe(A,G1,C,QE,RE);
26 PHIE = A-M*C*A;
27 GAMMAE = B-M*C*B;
28
29 %% Design of a simple first-order pre-filter
30 af = 0.9;
31 Afilt = [1 -af]; %Filter Denominator
32 Bfilt = 1-af; %Filter Numerator
33
34 %% Integral gain
35 K_I = 0.02;
```

Listing 1: LQR and LQG Design in MATLAB

The script mirrors the minimum set of actions that need to be executed prior to running experiments on the plant, because it defines all gains necessary to run the Simulink model described in the section 1.2 below. Besides this, and in order to save space for the report itself, code related to plots is only included in the attached MATLAB scripts.

1.2 Show and describe the block diagram used to control the system. (1.5 pts)

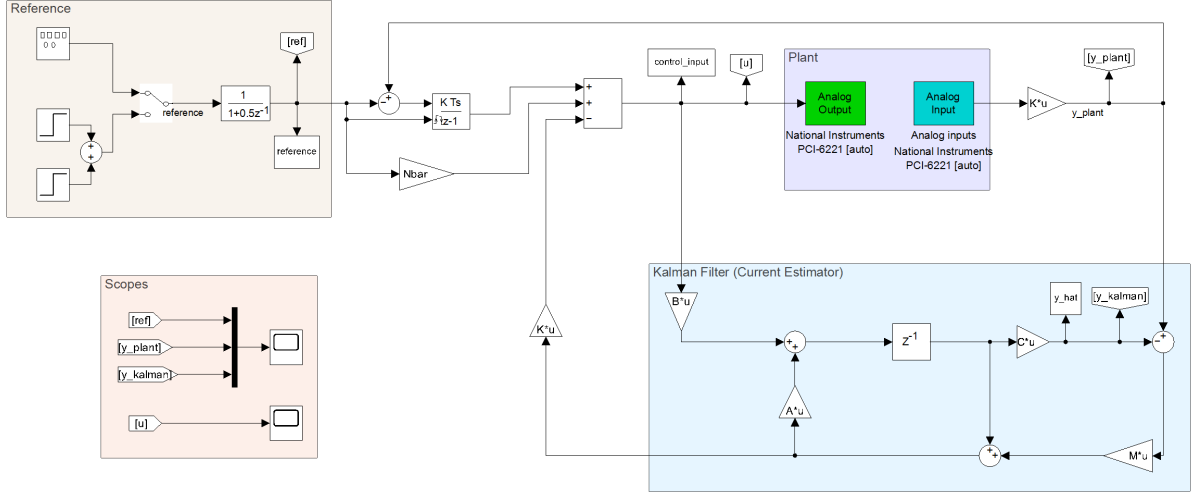


Figure 1: Simulink block diagram used to control the system

The block diagram was developed progressively, increasing in complexity at each stage. First, the identified discrete state-space model was used in place of the plant to design the Linear Quadratic Regulator (LQR) controller. Next, an observer was incorporated and used to feedback the estimated state to the plant, allowing for the design of a Linear Quadratic Estimator (LQE). Finally, to enable reference tracking, the reference block and compensation gain were computed, while the integral action on the error was implemented only in the physical setup. The complete block diagram is shown in figure ??.

The plant, highlighted in violet, functions as the interface between the control system and the physical setup. In practice, it converts the digital control input into an analog signal, applies the corresponding voltage to the motor that drives the flexible link, and converts the analog signals from the rotational potentiometer and strain gauge back into digital signals. In reality, this block represents not the physical plant itself, but rather the communication and signal exchange between the controller and the plant.

The observer, highlighted in blue, corresponds to a current Kalman filter. In short, it takes the same control input as the plant and, based on the plant model identified in the first report, predicts the state of the system. Then, this state is corrected according to the comparison of the measured bar deflection angle and the estimated deflection from the observer. Mathematically, the Kalman filter block executes the state prediction and correction loop described by equations 1 and 2

$$\hat{x}[n|n-1] = A\hat{x}[n-1|n-1] + Bu[n-1] + Gw[n-1] \quad (1)$$

$$\hat{x}[n|n] = \hat{x}[n|n-1] + M_x(y[n] - C\hat{x}[n|n-1] - Du[n]) \quad (2)$$

The reference block, highlighted in brown, enables the selection of different input signals for the plant. Additionally, a dedicated scope block facilitates real-time monitoring and visualization of these signals.

Putting this all together, the Kalman filter, which estimates the state of the system, feeds it back for linear state feedback, similar to a Linear Quadratic Regulator(LQR)

feedback. Because of this closed-loop control, it becomes necessary to introduce a compensation gain in the reference signal, in order to design a system with unitary gain.

As mentioned before, the integral of the error was only implemented in the physical setup control loop. It was observed the real motor had a significant dead zone, spanning from about -0.5 V to 0.5 V, motivating the inclusion of an integrator that resets according to the reference signal. In the simulation environment, this effect could also be reproduced, and thus make the integral action also necessary, introducing a "Dead Zone" block in Simulink.

Ultimately, the resulting block diagram establishes the necessary framework for a Linear Quadratic Gaussian (LQG) controller.

1.3 Comment on the choice of the weights on the quadratic cost when using the LQG design approach. Include the root-square locus. (3 pts)

The LQG controller combines the optimal state feedback control law (LQR) with a Kalman observer. The feedback gain K is obtained by minimizing the quadratic cost functional

$$J = \sum_{k=0}^{\infty} (x_k^T Q x_k + u_k^T R u_k) \quad (3)$$

where the matrix Q penalizes state errors and R penalizes the control effort. By adjusting these weights, it is possible to balance how fast and how smooth the system response will be.

In this project, the matrix Q was chosen as $Q = C^T C$, which gives more importance to minimizing the output error, and allows to tune the optimal feedback gain by changing the R matrix only. The parameter R was then varied to study its influence on performance.

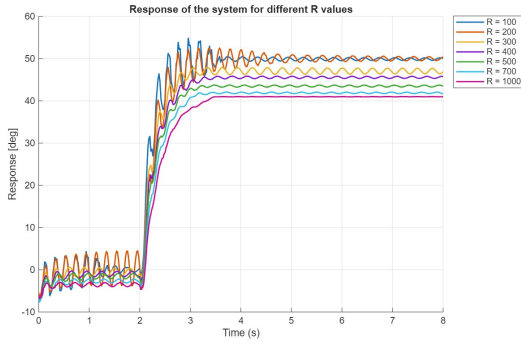


Figure 2: Response of the system for different R values

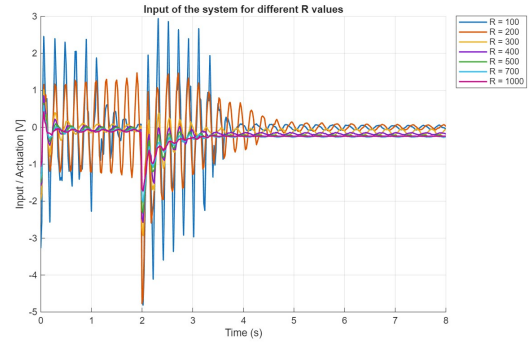


Figure 3: Input of the system for different R values

When R is small, the controller reacts faster but uses higher control inputs, which may saturate the actuator. When R is large, the control input becomes smoother and smaller, but the system reacts more slowly.

After testing several values, it was inferred $R = 400$ provided a good balance between response speed, stability, and control effort. This value ensures a stable and well-damped response without actuator saturation. Obviously, the requested control input also depends on the reference, as higher references are prone to cause saturation more easily.

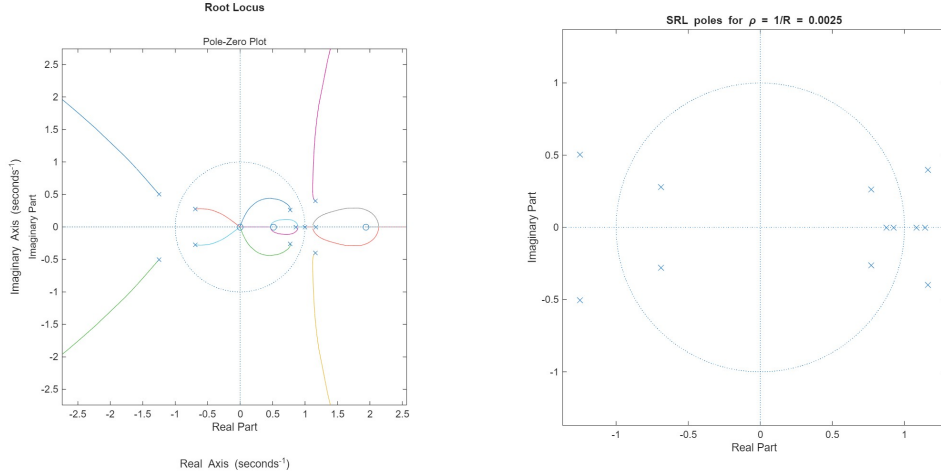


Figure 4: Root Locus (left) and pole-zero map (right) for $\rho = 1/R = 0.0025$.

All poles lie inside the unit circle, confirming a stable and well-damped response for $R = 400$. The *root-square locus* (SRL) illustrates the location of the closed-loop poles as R changes. The figure also shows the pole location for the chosen control weight $R = 400$ ($\rho = 1/R = 0.0025$). As the feedback gain of the closed-loop control increases the poles of the controlled system approach the zeros of the open-loop. Since there is a non-minimum phase zero, the SRL has to be used instead of the standard *root-locus*, and the Chang/Letov theorem guarantees that the closed-loop poles are the stable poles of the SRL. In general, the SRL is used to analyze how the poles move when R changes: for very small R , the poles move closer to the zeros, producing a faster and more oscillatory response; for very large R , the poles approach the open-loop poles, resulting in a slower but smoother system. With $R = 400$, the poles remain inside the unit circle and close to the real axis, which provides the desired compromise between speed and stability.

In short, the selected weights

$$Q = C^T C, \quad R = 400$$

give a good balance between accuracy, control effort, and robustness. The root-square locus confirms that the LQR controller is stable and properly tuned for this system.

1.4 Explain the effect of the choice of the noise covariance matrices in a LTR framework; (3 pts)

In discrete time, the LQR controller usually provides good stability margins, which are reduced by the inclusion of a Kalman filter. Therefore, the Loop Transfer Recovery (LTR) focuses on adjusting the covariance matrices of the LQE cost functional, Q_E and R_E , the process and measurement noise covariances, respectively, in a way that allows the open-loop frequency response of the LQG similar to that of the LQR. As with the LQR design, its the relative weight between both costs rather than their absolute value that matter.

In the frequency domain, this analysis was done with the identified model instead of the real plant. The resulting bode diagrams of the LQR and the LQG for different weights is shown in figure 5.

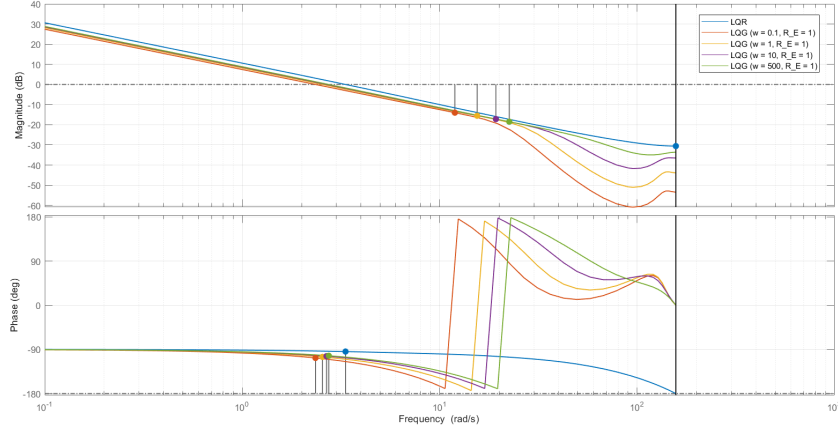


Figure 5: Open-loop bode plot of the LQR and LQG systems

As expected, the LQR has the best stability margins amongst all: it only reaches -180° when the frequency reaches the Nyquist frequency, and has a phase margin of almost 90° . Additionally, increasing the w value, thus the entries of Q_E , relative to the R_E increases the gain margin and the phase margin of the LQG system. In the lower frequencies, the gain is similar between all systems, but at the higher-end, in contrast to the stability margins, an increase in w reduces sensor noise attenuation and robust stability.

In other words, increasing the value of Q_E relative to R_E leads to a larger observer gain, therefore increasing sensor noise sensibility. On the other side, decreasing Q_E relative to R_E makes the state estimation less sensitive to sensor measurements, which makes the poles of the observer slower. This is not necessarily a downside as long as the estimator poles are faster than the controller ones.

Given that all the systems provide good stability margins, the optimal approach is to tune the weights such that the system is fast enough to provide on-time state estimation to the controller, and still attenuates sensor noise.

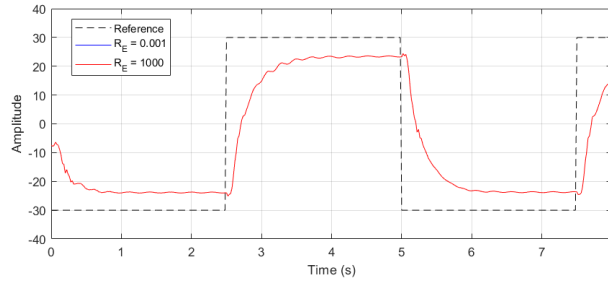


Figure 6: Plant's time response for different R_E values

From figure 6 above, it is unclear the real effect of changing the relative weights of the covariance matrices. With the decrease in R_E it would be expected that the system's response would become more oscillatory and noisy. Since that didn't happen, it was probably due to the fact that R_E was changed but the observer gains were not re-calculated afterwards. Nonetheless, there were also some non saved experiences where changing either matrix had the expected consequences in the system, visualized in the scopes of Simulink. Lastly, note that the signal does not follow the reference because at this point there was still no integrator added to the controller.

In the end, it was concluded that a value of $R_E = 1$ and $Q_E = wI$, with $w = 100$, provided good results.

1.5 Display and analyze the resulting closed-loop frequency response and time response. (3 pts)

The closed-loop system is described by the state-space:

$$A_{cl} = \begin{bmatrix} A & -BK \\ MCA & A - BK - MCA \end{bmatrix}, \quad B_{cl} = \begin{bmatrix} B \\ MCB \end{bmatrix}, \quad N_{bar}, C_{cl} = \begin{bmatrix} C & 0 \end{bmatrix}, \quad D_{cl} = 0,$$

and its frequency response is shown in Figure 7.

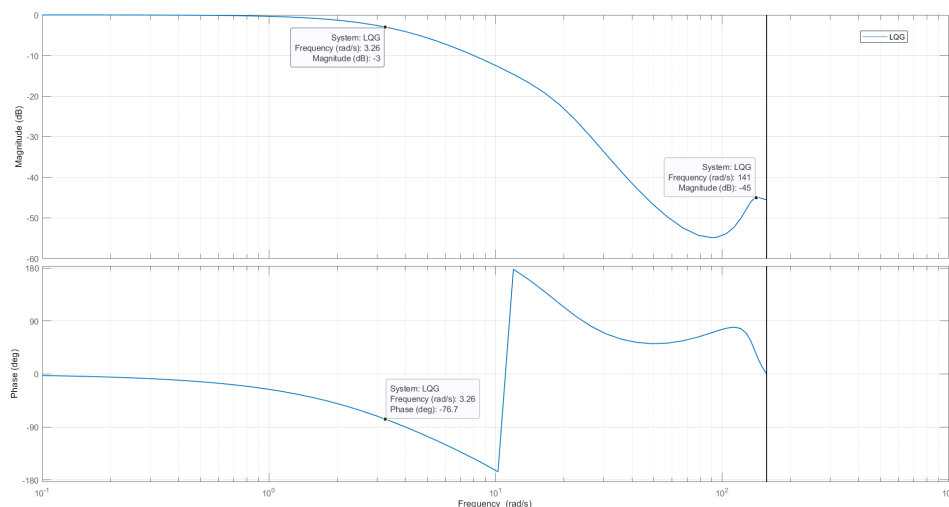


Figure 7: Bode plot of closed-loop system

At low frequencies, the gain is approximately 0dB and the bandwidth is 3.26 rad/s (≈ 0.52 Hz), meaning the system can track reference signals while maintaining a moderate response time. The frequency response exhibits a steep gain roll-off, demonstrating strong attenuation of high-frequency noise. A slight increase near the Nyquist frequency suggests the presence of a resonant frequency that may not be accurately captured due to the discretization of the system.

On top of this, the phase starts at 0°, highlighting the system's ability to track slow input signals with minimal lag. As the frequency increases, the phase decreases, indicating growing lag in the response. Particularly, at the bandwidth frequency, the lag is -76.7°: if the input to the system is a sine wave at that frequency, the system will react with almost a lag of a quarter period.

Overall, the frequency response of the closed-loop system suggests that, despite the somewhat limited bandwidth, it succeeds in tracking references and attenuating noise.

Figure 8 shows the closed-loop step response of the LQG controller. The system reacts quickly to the unit-step reference, reaching the final value in about 1–2 s with negligible overshoot. The steady-state error is practically zero due to the feedforward gain N_{bar} . The smooth shape of the curve indicates that the response is well-damped and consistent with the 3.26 rad/s bandwidth obtained from the frequency response. Overall, the time response confirms that the controller provides fast tracking and stability without excessive control effort.

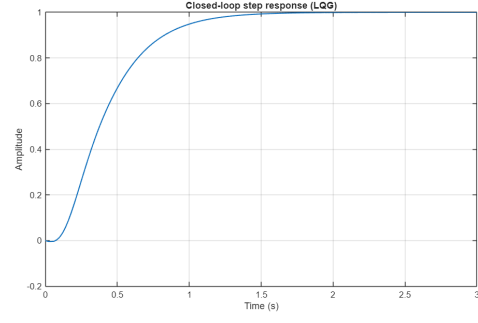


Figure 8: Closed-loop step response of the LQG controller

1.6 Comment on effect of the inclusion of a pre-filter. (2 pts)

When we generate a reference as a step or square wave, the system has to react to instantaneous variations of this reference. The motivation to include a pre-filter is to smoothen these reference variations, in order to get a slower but smoother system response. We introduced a low pass filter 4 and experimented with different values for the pole a_f

$$H(z) = \frac{Y_f(z)}{X(z)} = \frac{1 - a_f}{1 - a_f z^{-1}} \quad (4)$$

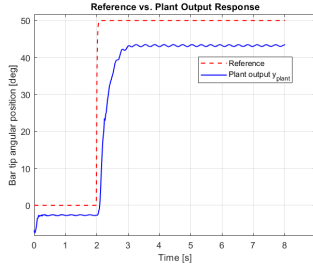


Figure 9: $a_f = 0.35$

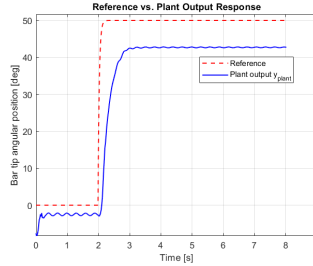


Figure 10: $a_f = 0.6$

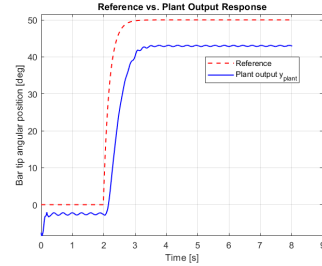


Figure 11: $a_f = 0.9$

As mentioned before, the dead zone of the motor causes a Steady-State error, with the plant never reaching the reference value. These experiments were done before the inclusion of an integrator. For that reason, we considered the reference as the value which the output converges to (43°), for the following calculations.

Metrics/ a_f	$a_f = 0.3$	$a_f = 0.6$	$a_f = 0.9$
Rise Time (s)	0.48396	0.47899	0.66376
Overshoot (%)	0.75049	0.28357	0.42026
Settling Time (s)	2.9386	2.8848	3.2018
RMSE	24.1084	23.9108	24.6733

Table 1: System Performance Metrics for Different a_f Values

Higher a_f values, mean lower cutoff frequencies, which smoothen the reference and, consequently, smoothen the actuation. However, there's a trade-off between a smooth actuation and low RMSE and Settling Times.

1.7 Discuss how do you evaluate the performance of your control system and what are the limits of performance. (3 pts)

To evaluate the performance of our control system, we used a series time-domain performance metrics: rise time, overshoot, settling time and RMSE. We also tested the system response to a square wave input.

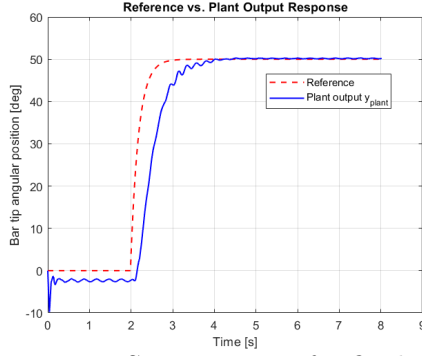


Figure 12: Step response for final controller parameters

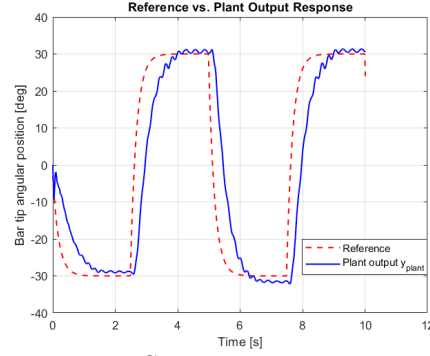


Figure 13: Square wave response for final controller parameters

Rise Time (s)	Overshoot (%)	Settling Time (s)	RMSE
0.84195	0.60205	3.729	7.3161

Table 2: System Performance Metrics for step input

While the Rise and Settling Times could be faster, we focused on having a stable system with minimal oscillations and unexaggerated control effort. The overshoot is minimal and we were able to cancel out the non minimum phase behavior of the system. Finally, the integrator significantly corrected the steady state error caused by the motor dead zone.

When tuning the system there were limitations and trade-offs that we had to take into account: The actuation is limited by the DC motor voltage range $[-10, 10](V)$. This limits how much the LQR R value can be lowered, as the actuator eventually saturates; The sensor noises limit how fast the estimation by Kalman Filter can be.

There are limits for the system performance: Input reference angles outside the $[-180, 180](^\circ)$ range cause problems, because of the non-linear zone of the potentiometer and tension of the cable connected to the strain gauge; The sampling frequency restricts the fastest stable response that can be achieved.

Conclusions

From this laboratory work, we gained a complete understanding of the control design cycle. This process heavily relies on the previous work of modeling the system. Tuning the controller involved understanding trade-offs, such as fast estimation against noise sensitivity and speed against control effort. Overall, we ended up with a totally different system response than what we saw when testing in open loop, demonstrating the impact a controller can have, even on such a simple system.